

1. Queries - Retrieve all Students

```
1 • CREATE DATABASE Student;
2 • USE Student;
3 • CREATE TABLE Student (
4     Student_ID INT,
5     Name VARCHAR(50),
6     Department VARCHAR(20),
7     Marks INT
8 );
9 • INSERT INTO Student VALUES
10 (101, 'Manu', 'SOCSE', 90),
11 (102, 'Varun', 'SOCSE', 80),
12 (103, 'Vaibhav', 'SOCSE', 70),
13 (104, 'Raghavendra', 'SOCSE', 60);
14 • SELECT *
15 FROM Student;
```

Output

Result Grid  Filter Rows: Export: 				
	Student_ID	Name	Department	Marks
▶	101	Manu	SOCSE	90
	102	Varun	SOCSE	80
	103	Vaibhav	SOCSE	70
	104	Raghavendra	SOCSE	60

2. Retrieve Selected Columns

```

1 • CREATE DATABASE CollegeDB;
2 • USE CollegeDB;
3 • CREATE TABLE Student (
4     Student_ID INT,
5     Name VARCHAR(50),
6     Department VARCHAR(20),
7     Marks INT
8 );
9 • INSERT INTO Student VALUES
10 (101, 'Manu', 'SOCSE', 90),
11 (102, 'Varun', 'SOCSE', 80),
12 (103, 'Vaibhav', 'SOCSE', 70),
13 (104, 'Raghavendra', 'SOCSE', 60);
14 • SELECT Name, Department, Marks
15 FROM Student;

```

Result Grid				Filter Rows:		Export:	Wrap Cell Content:
	Name	Department	Marks				
▶	Manu	SOCSE	90				
	Varun	SOCSE	80				
	Vaibhav	SOCSE	70				
	Raghavendra	SOCSE	60				

3. Find students scoring more than 80

SQL File 1* SQL File 2* SQL File 3* x

Limit to 1000 rows

```
1 • CREATE DATABASE CollegeAB;
2 • USE CollegeAB;
3 • CREATE TABLE Student (
4     Student_ID INT,
5     Name VARCHAR(50),
6     Department VARCHAR(20),
7     Marks INT
8 );
9 • INSERT INTO Student VALUES
10 (101, 'Manu', 'SOCSE', 90),
11 (102, 'Varun', 'SOCSE', 80),
12 (103, 'Vaibhav', 'SOCSE', 70),
13 (104, 'Raghavendra', 'SOCSE', 60);
14 • SELECT Name, Marks
15 FROM Student
16 WHERE Marks > 80;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

	Name	Marks
▶	Manu	90

4. Find CSE students

SQL File 1* SQL File 2* SQL File 3* SQL File 4* x

Limit to 1000 rows

```
1 • CREATE DATABASE CollegeEB;
2 • USE CollegeEB;
3 • CREATE TABLE Student (
4     Student_ID INT,
5     Name VARCHAR(50),
6     Department VARCHAR(20),
7     Marks INT
8 );
9 • INSERT INTO Student VALUES
10 (101, 'Manu', 'SOCSE', 90),
11 (102, 'Varun', 'SOCSE', 80),
12 (103, 'Vaibhav', 'SOB', 70),
13 (104, 'Raghavendra', 'SOCSE', 60);
14 • SELECT Name, Marks
15 FROM Student
16 WHERE Department = 'SOCSE';
```

Result Grid

	Name	Marks
▶	Manu	90
	Varun	80
	Raghavendra	60

Student 3 x

5. Find students from Delhi who scored above 80

SQL File 1* SQL File 2* SQL File 3* SQL File 4* SQL File 5* x

Limit to 1000 rows

```
1 • CREATE DATABASE CollegeHB;
2 • USE CollegeHB;
3 • CREATE TABLE Student (
4     Student_ID INT,
5     Name VARCHAR(50),
6     Department VARCHAR(20),
7     Marks INT,
8     City VARCHAR(50)
9 );
10 • INSERT INTO Student VALUES
11     (101, 'Manu', 'SOCSE', 90, 'Bengaluru'),
12     (102, 'Varun', 'SOCSE', 80, 'Bengaluru'),
13     (103, 'Vaibhav', 'SOB', 70, 'Bengaluru'),
14     (104, 'Raghavendra', 'SOCSE', 60, 'Delhi');
15 • SELECT Name, Marks
16 FROM Student
17 WHERE City = 'Delhi';
```

Result Grid Filter Rows: Export: Wrap Cell Content: [FA](#)

	Name	Marks
▶	Raghavendra	60

6. Find the highest marks

SQL File 1* SQL File 2* SQL File 3* SQL File 4* SQL File 5* SQL File 6* x

Limit to 1000 rows

```
1 • CREATE DATABASE CollegeIB;
2 • USE CollegeIB;
3 • CREATE TABLE Student (
4     Student_ID INT,
5     Name VARCHAR(50),
6     Department VARCHAR(20),
7     Marks INT,
8     City VARCHAR(50)
9 );
10 • INSERT INTO Student VALUES
11     (101, 'Manu', 'SOCSE', 90, 'Bengaluru'),
12     (102, 'Varun', 'SOCSE', 80, 'Bengaluru'),
13     (103, 'Vaibhav', 'SOB', 70, 'Bengaluru'),
14     (104, 'Raghavendra', 'SOCSE', 60, 'Delhi');
15 • SELECT MAX(Marks) AS Highest_Marks
16 FROM Student;
```

Result Grid

Highest_Marks
90

7. Find the average marks

The screenshot shows a SQL IDE with multiple tabs. The active tab, 'SQL File 7*', contains the following SQL script:

```
1 • CREATE DATABASE CollegeJB;
2 • USE CollegeJB;
3 • CREATE TABLE Student (
4     Student_ID INT,
5     Name VARCHAR(50),
6     Department VARCHAR(20),
7     Marks INT,
8     City VARCHAR(50)
9 );
10 • INSERT INTO Student VALUES
11     (101, 'Manu', 'SOCSE', 90, 'Bengaluru'),
12     (102, 'Varun', 'SOCSE', 80, 'Bengaluru'),
13     (103, 'Vaibhav', 'SOB', 70, 'Bengaluru'),
14     (104, 'Raghavendra', 'SOCSE', 60, 'Delhi');
15 • SELECT AVG(Marks) AS Average_Marks
16 FROM Student;
```

Below the script editor, the 'Result Grid' is visible, showing the output of the last query:

Average_Marks
75.0000

On the right side of the IDE, a vertical sidebar contains the text 'Auditing current'.

Subqueries

1. Students scoring above the average

The screenshot shows a SQL IDE with multiple tabs. The active tab, 'SQL File 8*', contains the following SQL code:

```
3 CREATE TABLE Student (  
4     Student_ID INT,  
5     Name VARCHAR(50),  
6     Department VARCHAR(20),  
7     Marks INT,  
8     City VARCHAR(50)  
9 );  
10 INSERT INTO Student VALUES  
11 (101, 'Manu', 'SOCSE', 90, 'Bengaluru'),  
12 (102, 'Varun', 'SOCSE', 80, 'Bengaluru'),  
13 (103, 'Vaibhav', 'SOB', 70, 'Bengaluru'),  
14 (104, 'Raghavendra', 'SOCSE', 60, 'Delhi');  
15 SELECT Name, Marks  
16 FROM Student  
17 WHERE Marks > (  
18     SELECT AVG(Marks)  
19     FROM Student  
20 );
```

Below the code editor, the 'Result Grid' is visible, showing the results of the query. It has columns 'Name' and 'Marks'.

Name	Marks
Manu	90
Varun	80

2. Student who scored the highest marks

The screenshot shows a SQL IDE with multiple tabs. The active tab is 'SQL File 9*'. The query editor contains the following SQL code:

```
3 • CREATE TABLE Student (  
4     Student_ID INT,  
5     Name VARCHAR(50),  
6     Department VARCHAR(20),  
7     Marks INT,  
8     City VARCHAR(50)  
9 );  
10 • INSERT INTO Student VALUES  
11 (101, 'Manu', 'SOCSE', 90, 'Bengaluru'),  
12 (102, 'Varun', 'SOCSE', 80, 'Bengaluru'),  
13 (103, 'Vaibhav', 'SOB', 70, 'Bengaluru'),  
14 (104, 'Raghavendra', 'SOCSE', 60, 'Delhi');  
15 • SELECT Name, Marks  
16 FROM Student  
17 WHERE Marks = (  
18     SELECT MAX(Marks)  
19 FROM Student  
20 );
```

Below the query editor, there is a 'Result Grid' section. It includes a 'Filter Rows:' field, an 'Export:' button, and a 'Wrap Cell Content:' checkbox. The result grid itself shows the following data:

Name	Marks
Manu	90

On the right side of the IDE, there is a blue button labeled 'Result Grid'.

3. Students scoring above

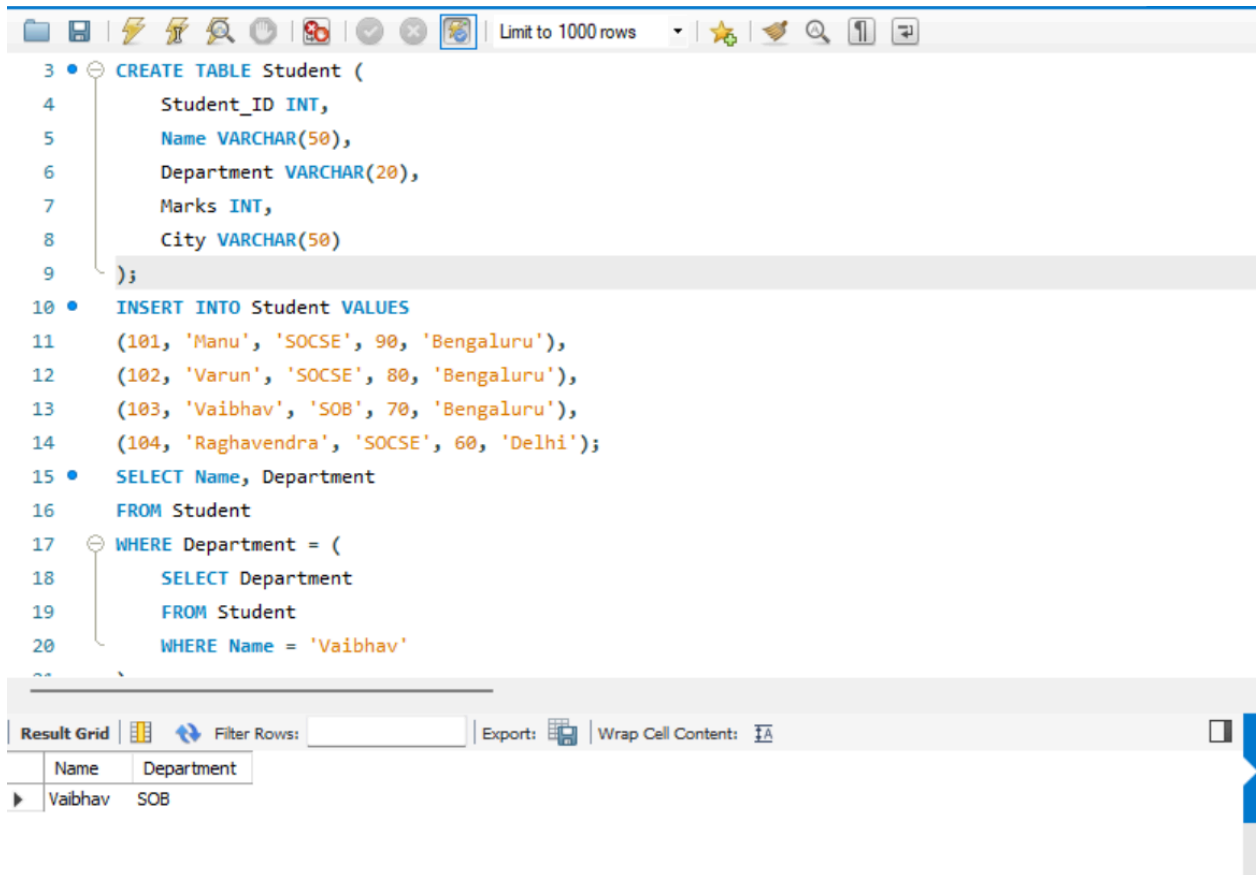
The screenshot shows a SQL IDE with a query editor containing the following SQL code:

```
3 CREATE TABLE Student (  
4     Student_ID INT,  
5     Name VARCHAR(50),  
6     Department VARCHAR(20),  
7     Marks INT,  
8     City VARCHAR(50)  
9 );  
10 INSERT INTO Student VALUES  
11 (101, 'Manu', 'SOCSE', 90, 'Bengaluru'),  
12 (102, 'Varun', 'SOCSE', 80, 'Bengaluru'),  
13 (103, 'Vaibhav', 'SOB', 70, 'Bengaluru'),  
14 (104, 'Raghavendra', 'SOCSE', 60, 'Delhi');  
15 SELECT Name, Marks  
16 FROM Student  
17 WHERE Marks > (  
18     SELECT Marks  
19     FROM Student  
20     WHERE Name = 'Vaibhav')
```

Below the query editor, the 'Result Grid' is visible, showing the results of the query. The grid has two columns: 'Name' and 'Marks'. The results are:

Name	Marks
Manu	90
Varun	80

4. Students belonging to the same department as



The screenshot shows a SQL IDE interface. The query editor contains the following SQL code:

```
3 • CREATE TABLE Student (  
4     Student_ID INT,  
5     Name VARCHAR(50),  
6     Department VARCHAR(20),  
7     Marks INT,  
8     City VARCHAR(50)  
9 );  
10 • INSERT INTO Student VALUES  
11     (101, 'Manu', 'SOCSE', 90, 'Bengaluru'),  
12     (102, 'Varun', 'SOCSE', 80, 'Bengaluru'),  
13     (103, 'Vaibhav', 'SOB', 70, 'Bengaluru'),  
14     (104, 'Raghavendra', 'SOCSE', 60, 'Delhi');  
15 • SELECT Name, Department  
16 FROM Student  
17 WHERE Department = (  
18     SELECT Department  
19 FROM Student  
20 WHERE Name = 'Vaibhav'
```

Below the query editor, the result grid is displayed with the following data:

Name	Department
Vaibhav	SOB

5. Students who scored above the average CSE marks

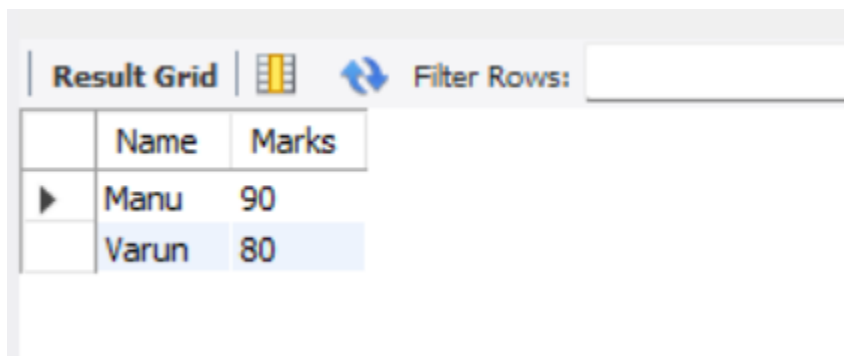
```
-> CREATE DATABASE CollegeOB;  
USE CollegeOB;  
CREATE TABLE Student (  
    Student_ID INT,  
    Name VARCHAR(50),  
    Department VARCHAR(20),  
    Marks INT,  
    City VARCHAR(50)  
);  
INSERT INTO Student VALUES  
(101, 'Manu', 'SOCSE', 90, 'Bengaluru'),  
(102, 'Varun', 'SOCSE', 80, 'Bengaluru'),  
(103, 'Vaibhav', 'SOB', 70, 'Bengaluru'),  
(104, 'Raghavendra', 'SOCSE', 60, 'Delhi');
```

```

SELECT Name, Marks
FROM Student
WHERE Department = 'SOCSE'
AND Marks > (
    SELECT AVG(Marks)
    FROM Student
    WHERE Department = 'SOCSE'
);

```

OUTPUT



The screenshot shows a 'Result Grid' window with a toolbar at the top containing icons for grid view, refresh, and a 'Filter Rows' input field. Below the toolbar is a table with two columns: 'Name' and 'Marks'. The table contains two rows of data: 'Manu' with a mark of 90, and 'Varun' with a mark of 80. The 'Varun' row is highlighted in blue.

	Name	Marks
▶	Manu	90
	Varun	80

6. Students who belong to a department having a student from Delhi

```

→ CREATE DATABASE CollegePB;
USE CollegePB;
CREATE TABLE Student (
    Student_ID INT,
    Name VARCHAR(50),
    Department VARCHAR(20),
    Marks INT,
    City VARCHAR(50)
);
INSERT INTO Student VALUES
(101, 'Manu', 'SOCSE', 90, 'Bengaluru'),
(102, 'Varun', 'SOCSE', 80, 'Bengaluru'),
(103, 'Vaibhav', 'SOB', 70, 'Bengaluru'),

```

```
(104, 'Raghavendra', 'SOCSE', 60, 'Delhi');
SELECT Name, Department
FROM Student
WHERE Department IN (
    SELECT Department
    FROM Student
    WHERE City = 'Delhi'
);
```

OUTPUT

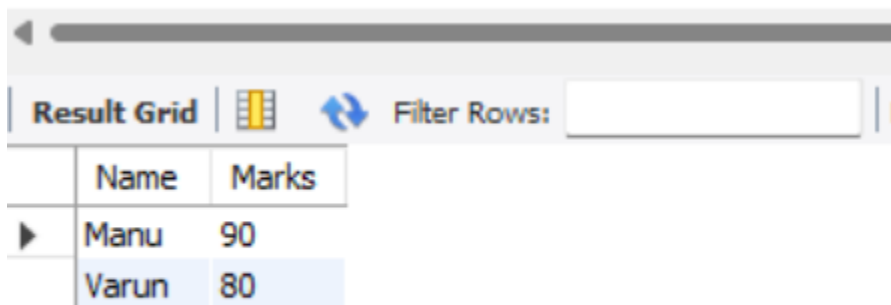
Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	Name	Department			
▶	Manu	SOCSE			
	Varun	SOCSE			
	Raghavendra	SOCSE			

7. Students who scored higher than every SOB student

```
CREATE DATABASE CollegeQB;
USE CollegeQB;
CREATE TABLE Student (
    Student_ID INT,
    Name VARCHAR(50),
    Department VARCHAR(20),
    Marks INT,
    City VARCHAR(50)
);
INSERT INTO Student VALUES
(101, 'Manu', 'SOCSE', 90, 'Bengaluru'),
(102, 'Varun', 'SOCSE', 80, 'Bengaluru'),
(103, 'Vaibhav', 'SOB', 70, 'Bengaluru'),
(104, 'Raghavendra', 'SOCSE', 60, 'Delhi');
```

```
SELECT Name, Marks
FROM Student
WHERE Marks > ALL (
    SELECT Marks
    FROM Student
    WHERE Department = 'SOB'
);
```

OUTPUT



The screenshot shows a database interface with a 'Result Grid' tab. The grid contains two columns: 'Name' and 'Marks'. There are two rows of data: 'Manu' with '90' marks and 'Varun' with '80' marks. The 'Varun' row is highlighted in blue. Above the grid, there is a 'Filter Rows:' input field and a refresh icon.

	Name	Marks
▶	Manu	90
	Varun	80